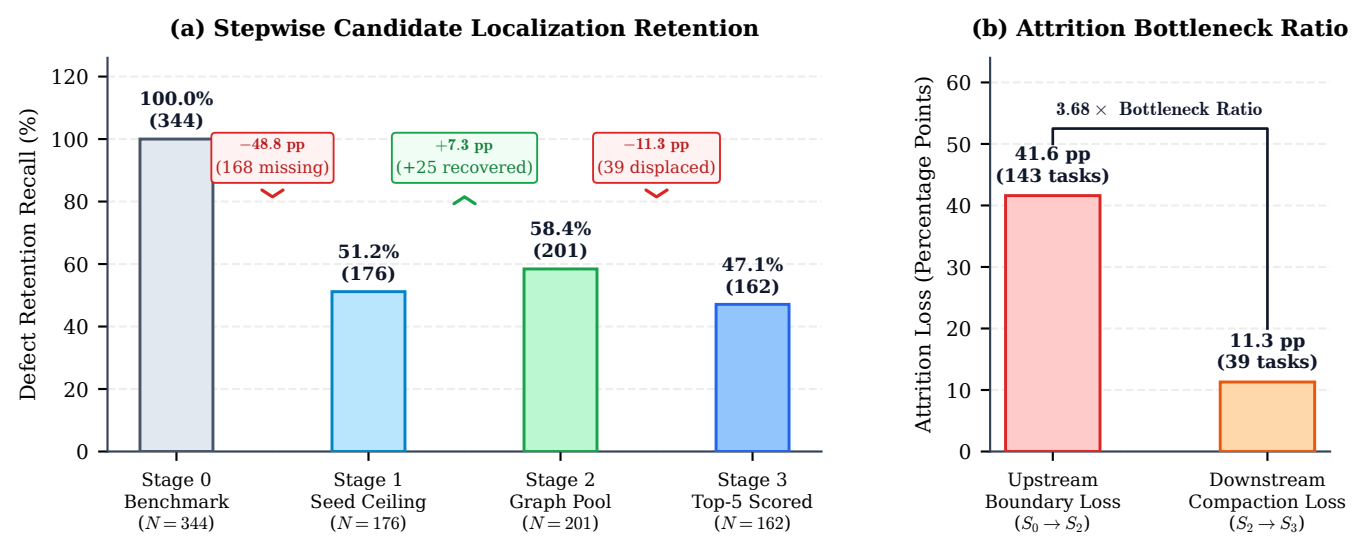
(a) Stepwise Candidate Localization Retention
Defect Retention Recall (%)
120
100
80
60
40
20
0
100.0%
(344)
−48.8 pp
(168 missing)
51.2%
(176)
+7.3 pp
(+25 recovered)
58.4%
(201)
−11.3 pp
(39 displaced)
47.1%
(162)
Stage 0
Benchmark
(N = 344)
Stage 1
Seed Ceiling
(N = 176)
Stage 2
Graph Pool
(N = 201)
Stage 3
Top-5 Scored
(N = 162)
(b) Attrition Bottleneck Ratio
Attrition Loss (Percentage Points)
60
50
40
30
20
10
0
3.68 × Bottleneck Ratio
41.6 pp
(143 tasks)
11.3 pp
(39 tasks)
Upstream
Boundary Loss
(S0 → S2)
Downstream
Compaction Loss
(S2 → S3)